

# E-Commerce Sales Analytics Dashboard

## DATA SCIENCE (24AD3107) Project Review-1



### Under the Guidance of

**Dr. Prathusha**

**Department of Computer Science and Engineering,  
KLH CSE Bowrampet Campus**



### Team Members

**Sri Ram -2420030763**

**Dinesh -2420030372**

**Venkat Ranjeeth-2420030569**

**Roll Number**

# Outline of Project Work

- Focus

- Introduction

- Literature Survey

- Challenges

- Research Objectives

- Conclusion & Future Work

- References

- Publications Details

# Problem Statement

- E-commerce generates a large amount of sales data.
- Raw data is difficult to analyze manually.
- Important sales and customer patterns may be difficult to identify.
- Businesses need an easy way to understand sales performance.
- Future sales are difficult to estimate from raw data.
- Therefore, we develop an E-Commerce Sales Analytics Dashboard using Data Science and Machine Learning.

# Objectives

- To analyze e-commerce sales data using Data Science techniques.
- To clean and preprocess the collected sales data.
- To identify sales trends, top-selling products, and profitable categories.
- To analyze customer and regional sales performance.
- To develop an interactive dashboard for easy data visualization.
- To use Machine Learning to predict future sales.
- To support data-driven business decision-making.

# Literature Survey

| Authors                                                                   | Title of the Paper                                                                          | Methodology                                                                                                                                          | Inferences                                                                                                                                                                     | Limitation / Future Work                                                                                                                                                                                                 |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chunlei Du, Yancai Li                                                     | E-commerce Sales Forecasting Based on Deep Learning Algorithm (2025)                        | <ul style="list-style-type: none"> <li>• Multi-layer LSTM</li> <li>• Temporal feature processing</li> <li>• Dropout and L2 regularization</li> </ul> | <ul style="list-style-type: none"> <li>• Forecasts future e-commerce sales using historical data.</li> <li>• Deep learning can capture complex sales patterns.</li> </ul>      | <ul style="list-style-type: none"> <li>• Focuses mainly on sales forecasting.</li> <li>• Future work can test the model on different e-commerce datasets and scenarios.</li> </ul> <a href="#">ScienceDirect</a>         |
| Mussadiq Abdul Rahim, Muhammad Mushafiq, Salabat Khan, Zulfiqar Ali Arain | RFM-Based Repurchase Behavior for Customer Classification and Segmentation (2021)           | <ul style="list-style-type: none"> <li>• RFM analysis</li> <li>• MLP</li> <li>• SVM</li> <li>• Decision Tree</li> </ul>                              | <ul style="list-style-type: none"> <li>• Identifies customer purchasing behavior.</li> <li>• RFM features can be used for customer classification and segmentation.</li> </ul> | <ul style="list-style-type: none"> <li>• Mainly focuses on repurchase behavior.</li> <li>• Future work can combine segmentation with sales forecasting and dashboard analytics.</li> </ul> <a href="#">ScienceDirect</a> |
| Naresh Kumar                                                              | Intelligent Customer Segmentation: Unveiling Consumer Patterns with Machine Learning (2025) | <ul style="list-style-type: none"> <li>• RFM analysis</li> <li>• Data preprocessing</li> <li>• Bisecting K-Means</li> <li>• Elbow method</li> </ul>  | <ul style="list-style-type: none"> <li>• Groups customers based on purchasing behavior.</li> <li>• Identifies different customer spending and activity patterns.</li> </ul>    | <ul style="list-style-type: none"> <li>• Focuses mainly on customer segmentation.</li> <li>• Future work can integrate segmentation with sales prediction and real-time analytics.</li> </ul> <a href="#">DOI System</a> |

# Problem Gap Identification

- Existing e-commerce systems generate large amounts of sales data but often provide limited combined analysis of products, customers, regions, and profits.
- Many existing approaches focus mainly on sales forecasting or customer segmentation separately.
- Raw sales data is difficult for business users to understand without proper visualization and dashboards.
- There is a need to combine data cleaning, exploratory analysis, visualization, customer segmentation, and sales prediction in one system.
- Existing models may be developed using specific datasets, making it necessary to test the approach on different e-commerce datasets.
- Therefore, our project aims to develop an integrated E-Commerce Sales Analytics Dashboard that provides historical insights and future sales predictions.

# Innovation, Creativity, and Novelty

- **Integrated Analytics:** Combines sales analysis, customer analysis, and profit analysis in a single dashboard.
- **Interactive Visualization:** Converts complex sales data into easy-to-understand charts, graphs, and tables.
- **Machine Learning Prediction:** Uses historical sales data to predict possible future sales.
- **Customer Segmentation:** Groups customers based on their purchasing behavior and spending patterns.
- **Data-Driven Insights:** Identifies top-selling products, profitable categories, high-performing regions, and monthly trends.
- **User-Friendly Dashboard:** Presents important business information in one place for quick understanding.
- **Scalable Approach:** The system can be extended with recommendation systems, real-time analytics, and advanced prediction models.

# Feasibility and Project Plan

## Feasibility

- **Technical:** Python, ML & dashboard tools.
- **Economic:** Low-cost, open-source tools.
- **Operational:** Easy-to-use dashboard.

## Timeline

**Data Collection → Data Cleaning → Analysis → Dashboard → ML  
Prediction → Testing**

**Duration:** 3 Months.